Nutritive Value of Foods

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Abstract

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This publication gives in tabular form the nutritive values for household measures of commonly used foods. It was first published in 1960; the last revision was published in 1991. In this revision, values for total dietary fiber have been added and phosphorus values have been removed. Values are reported for water; calories; protein; total fat; saturated, monounsaturated, and polyunsaturated fatty acids; cholesterol; carbohydrate; total dietary fiber; calcium; iron; potassium; sodium; vitamin A in IU and RE units; thiamin; riboflavin; niacin; and ascorbic acid (vitamin C). Data are from the U.S. Department of Agriculture Nutrient Database for Standard Reference, Release 13.

Keywords: ascorbic acid, calcium, calories, cholesterol, dietary fiber, fatty acids, foods, iron, niacin, nutrient composition, nutrient data, potassium, protein, riboflavin, salt, sodium, total fat, vitamin A

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Abbreviations

dia	diameter
fl oz	fluid ounce
g	gram
kcal	kilocalorie (commonly known as calories)
IU	International Units
lb	pound
μg	microgram
mg	milligram
ml	milliliter
NA	not available
oz	ounce
pkg	package
RE	retinol equivalent
sq	square
tbsp	tablespoon
Tr	trace
tsp	teaspoon

Nutritive Value of Foods

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Introduction

An 8-oz glass of milk, a 3-oz slice of cooked meat, an apple, a slice of bread. What food values does each contain? How much cooked meat will a pound of raw meat yield? How much protein should a healthy 14-year-old boy get each day?

Consumers want ready answers to questions like these so they can plan nutritious diets for themselves and their families. Also, nutritionists, dietitians, and other health professionals use this type of information in their daily work.

In response, the U.S. Department of Agriculture published the first edition of this bulletin in 1960. USDA nutrition researchers have revised it many times since to reflect our expanded knowledge, to add or subtract specific values, and to update the ever-growing list of available, commonly used foods.

Further Information

The USDA Nutrient Database for Standard Reference is a more technical compilation of nutrient information, with data for a much more extensive list of foods and nutrients than this publication provides. It is revised regularly and published on the USDA Nutrient Data Laboratory (NDL) web site, <www.nal.usda.gov/fnic/foodcomp>. It replaces USDA's Agriculture Handbook 8, "Composition of Foods. . .Raw, Processed, Prepared," commonly referred to as "Handbook 8," and its revised sections, which are out of print. Special-interest tables—such as Isoflavone Content of Foods—are also published on the NDL web site.

The USDA Nutrient Database for Standard Reference and special-interest tables produced by NDL are also available on CD-ROM from the U.S. Government Printing Office (GPO). See the back of the title page for contact information.

Other nutrition publications that may be useful include "Nutrition and Your Health: Dietary Guidelines for Americans," USDA Home and Garden Bulletin 232; "Making Healthy Food Choices," USDA Home and Garden Bulletin 250; and "Check It Out: The Food Label, the Pyramid, and You," USDA Home and Garden Bulletin 266. These publications may also be purchased from GPO. See the back of the title page for contact information.

The Dietary Guidelines for Americans and the Food Guide Pyramid can be found on USDA's Center for Nutrition Policy and Promotion web site, <http://www.usda.gov/cnpp>, or write to them at 3101 Park Center Dr., Room 1064, Alexandria, VA 22302-1594. Food label and other nutrition information can be found on the Food and Drug Administration's Center for Food Safety and Applied Nutrition web site, <http://vm.cfsan.fda.gov/label.html>, or write to them at 200 C Street, SW, Washington, DC 20204.

Full texts of the Recommended Dietary Allowances and each volume of Dietary Reference Intakes are available from the National Academy Press, at www.nap.edu or 888-624-8373 (toll free).

For more information about food and nutrition, visit the USDA-ARS National Agricultural Library's Food and Nutrition Information Center <http://www.nal.usda.gov/fnic/>, or contact them at 10301 Baltimore Ave., Room 304, Beltsville, MD 20705-2351, Phone: 301-504-5719, Fax: 301-504-6409, TTY: 301-504-6856, e-mail: fnic@nal.usda.gov. Another source of information on the Internet is <www.nutrition.gov>.

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Table 1. Equivalents by Volume and Weight

This table contains some helpful volume and weight equivalents. Following is an example that illustrates how you can use the table:

Example. For milk, the nutrient profile covers a 1-cup serving (see page 20, table 9). Let's say you use 2 tablespoons of milk in your coffee. In table 1, you see that 1 cup equals 16 tablespoons, so the 2 tablespoons you consume are two-sixteenths or one-eighth of 1 cup. To find out the nutritive value of the amount you actually consume—2 tablespoons—you need to divide the nutrient values listed for milk by 8.

Volume

1 gallon (3.786 liters; 3,786 ml)	4 quarts
1 quart (0.946 liter; 946 ml)	4 cups or 2 pints
1 cup (237 ml)	8 fluid ounces, ½ pint, or 16 tablespoons
2 tablespoons (30 ml)	1 fluid ounce
1 tablespoon (15 ml)	3 teaspoons
1 pint	2 cups

Weight

1 pound (16 ounces)	453.6 grams
1 ounce	28.35 grams
3½ ounces	100 grams

Table 2. Tips for Estimating Amount of Food Consumed

This table lists some handy tips to help you estimate the amount of food you eat when you cannot measure or weigh it.

Breads and grains	
½ cup cooked cereal, pasta, rice	volume of cupcake wrapper or half a baseball
4-oz bagel (large)	diameter of a compact disc (CD)
medium piece of cornbread	medium bar of soap
Fruits and vegetables	
medium apple, orange, peach	tennis ball
¼ cup dried fruit	golf ball or scant handful for average adult
½ cup fruit or vegetable	half a baseball
1 cup broccoli	light bulb
medium potato	computer mouse
1 cup raw leafy greens	baseball or fist of average adult
½ cup	6 asparagus spears, 7 or 8 baby carrots or carrot sticks, or a medium ear of corn
Meat, fish, and poultry, cooked	
1 oz	about 3 tbsp meat or poultry
2 oz	small chicken drumstick or thigh
3 oz	average deck of cards, palm of average adult's hand, half of a whole, small chicken breast, medium pork chop
Cheese	
1 oz hard cheese	average person's thumb, 2 dominoes, 4 dice
Other	
2 tbsp peanut butter	Ping-Pong ball
⅓ cup nuts	level handful for average adult
½ cup	half a baseball or base of computer mouse
1 cup	tennis ball or fist of average adult

Note: The serving size indicated in the Food Guide Pyramid and on food labels is a standardized unit of measure and may not represent the portion of food a person actually eats on one occasion.

Sources: Schuster (1997), American Institute of Cancer Research (2001).